

Machine Learning in Physics

Introduction · Lecture 1 of 2

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Slide design and drafting assisted by Claude (Anthropic)

About me

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CHIME/FRB collaboration



Research

1. **FRB populations**
   Volumetric rate density, energy and redshift distributions

2. **Microsecond structure in FRBs**
   Constraining FRB emitting engines and emission physics

3. **CHIME/FRB upgrade**
   Searching for more microsecond-scale transients

4. **CHIME/FRB data processing pipelines**
   Calibration, flux/fluence, and sensitivity measurements

Quick poll

How do you use machine learning in your research, or how might you?



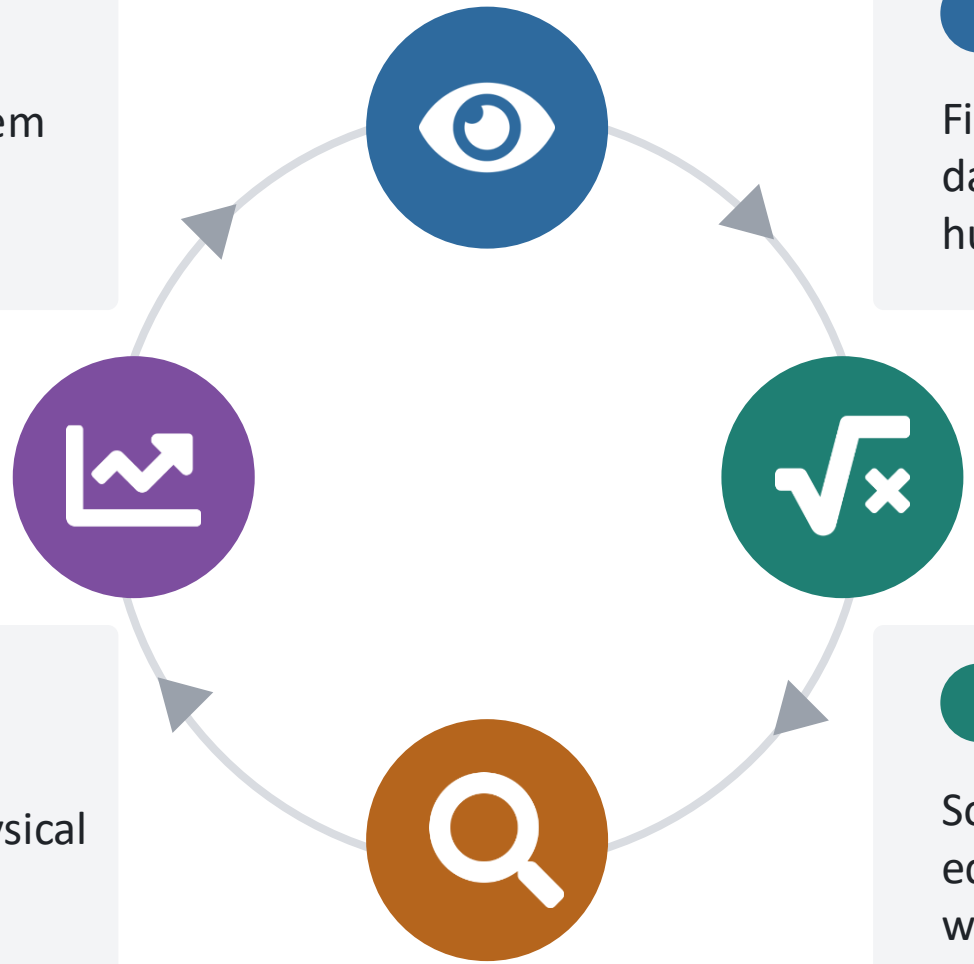
ML in Physics: Overview

Prediction

Forecast future states of a system or unlock discovery spaces

Observations

Filtering massive streams of raw data that are impossible for humans to sort manually



Inference

Figuring out the underlying physical laws or parameters resulting in observed data

Modeling

Solving complex differential equations which can take days or weeks on supercomputers

ML in Physics: Examples

Observations

1. Event filtering at **Large Hadron Collider**
2. Noise reduction and signal enhancement by **Event Horizon Telescope**
3. Anomaly detection in **astronomical surveys**

Modeling

1. **Emulating** complex simulations (e.g., fluid dynamics, climate models)
2. **Physics-Informed Neural Networks (PINNs)** to solve complex PDEs more efficiently than traditional numerical methods

Inference

1. Masses and spins of black holes in gravitational wave detections by **LIGO**
2. Phase transitions into new states of matter, such as **superconductors**

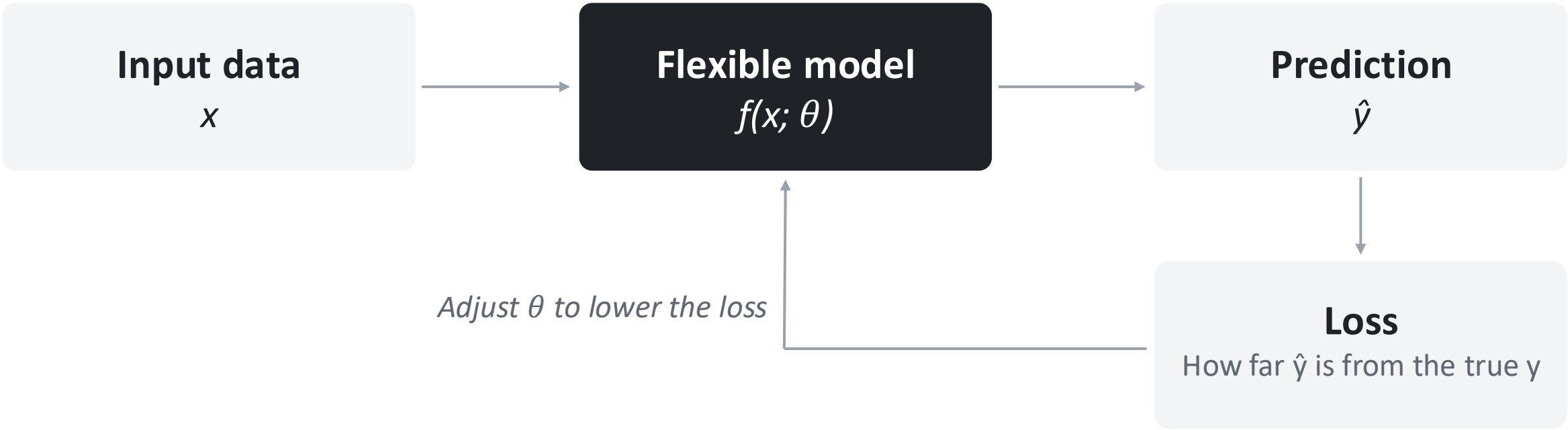
Prediction

1. **Weather and climate** forecasting
2. Predict the existence of and search for **exotic new particles**
3. Predict which **FRBs** are highly likely to repeat in the future

Sources: Albrecht et al., 2025, J. Phys. G: Nucl. Part. Phys. 52 030501 · Janssen et al., 2025, A&A, 698, A62 · Etsebeth et al., 2024, MNRAS, 529(1), 732-747 · Suresh R, et al., 2024, Front. Phys., 12:1322162 · George & Huerta, 2018, Physics Letters B, 778, 64-70.

Machine Learning

ML is the process of training a **model** to make useful **predictions** from **data**.



Sources: Google Machine Learning Education, Machine Learning Roadmap

The goal: predicting new data

Linear Regression / MCMC

Fit for parameters in a physical model, either minimizing errors or sampling the posterior

Machine learning

Fit for parameters while reducing loss, to make predictions without using an explicit model

GENERAL IDEA



1. Train the model on a robust data set (real or simulated).
2. Test its predictive power on held-out data (real or simulated) it never saw during training.
3. Apply to new data.

Sources: Google Machine Learning Education, Machine Learning Roadmap

Supervised and unsupervised learning



Supervised

Data provides labeled examples, i.e., inputs with known answers. The model learns the mapping from one to the other.

TASKS

Regression | Classification



Unsupervised

There are no labels. The model looks for structure in the data on its own.

TASKS

Clustering | Dimensionality reduction

Sources: Google Machine Learning Education, Machine Learning Roadmap

Regression

Supervised **Inference**

Find a relationship between features and a continuous label

COMMON METHODS

- Linear and ridge regression
- Random forests
- Gradient boosting
- Neural Networks

CAVEATS

- Point estimates, usually without uncertainties
- Unreliable beyond the training data's range

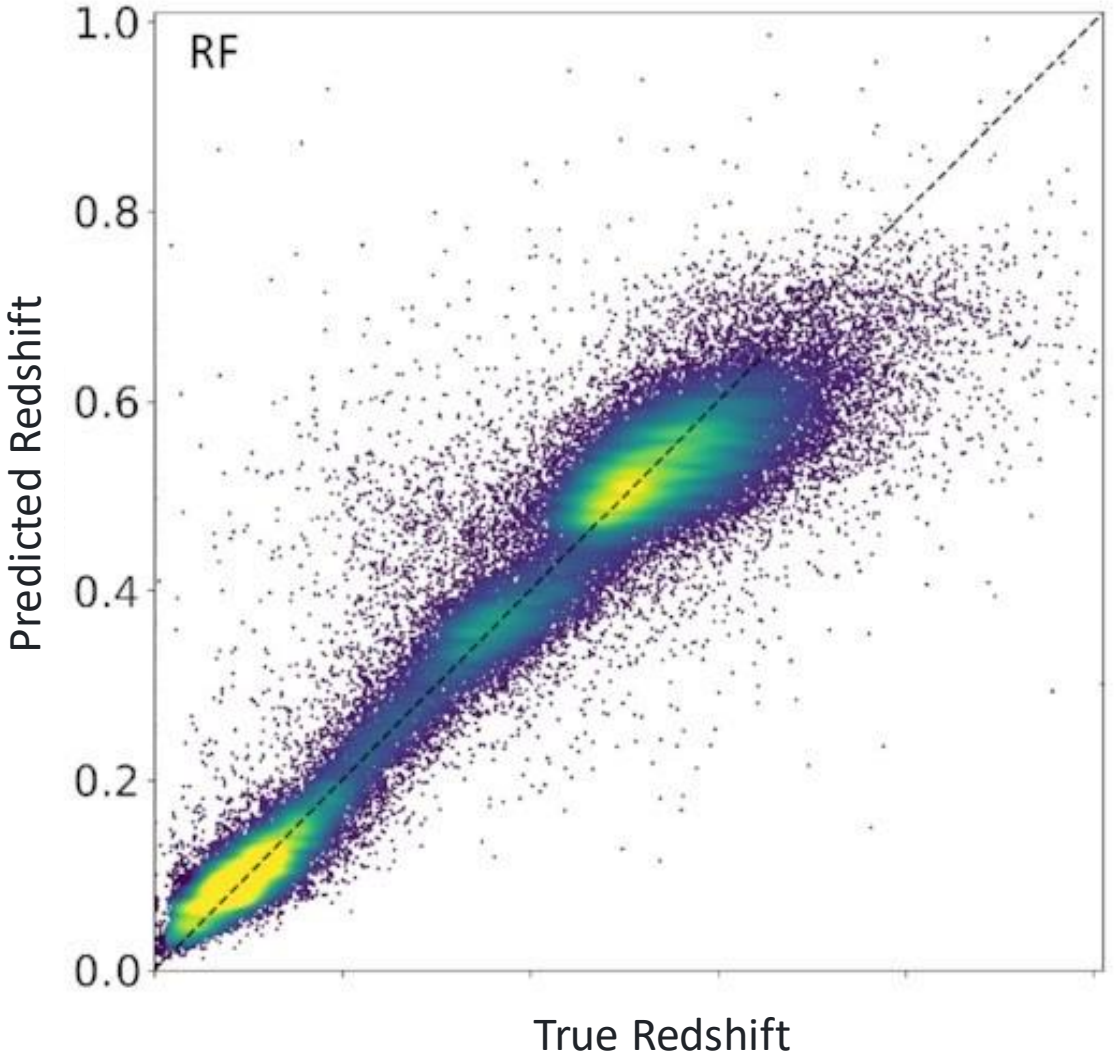


Figure from: Henghes, B. et al., 2022, MNRAS, 512(2), 1696-1709.

Classification

Supervised **Observations**

Assign each object to one of a set of known categories

COMMON METHODS

- Logistic regression
- Random forests
- Nearest neighbors
- Neural Networks

CAVEATS

- Can only assign classes it was trained on
- Inherits any biases in its training labels

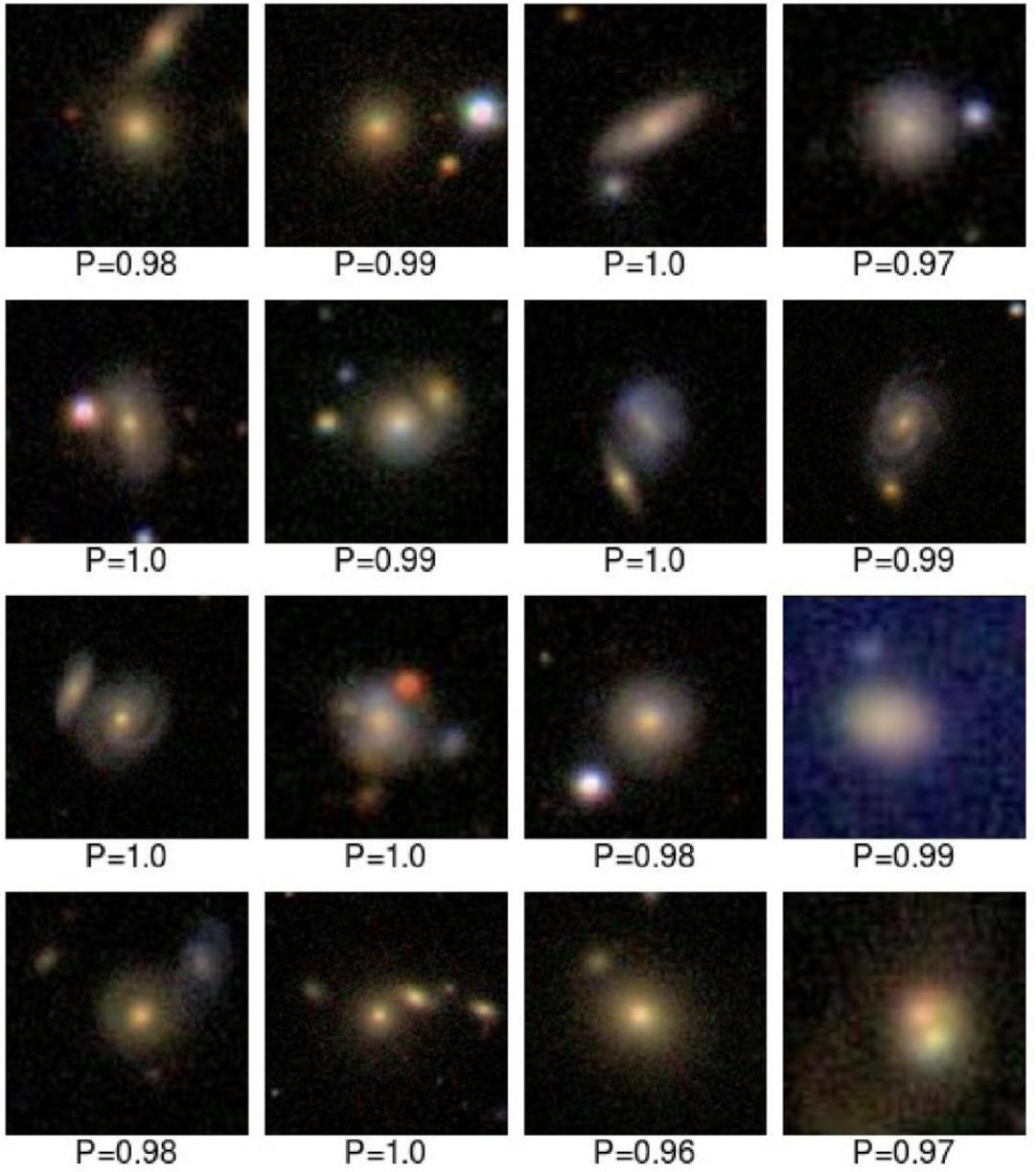


Figure from: Domínguez Sánchez, H. et al., 2018, MNRAS, 476(3), 3661-3676.

Clustering

Unsupervised **Observations**

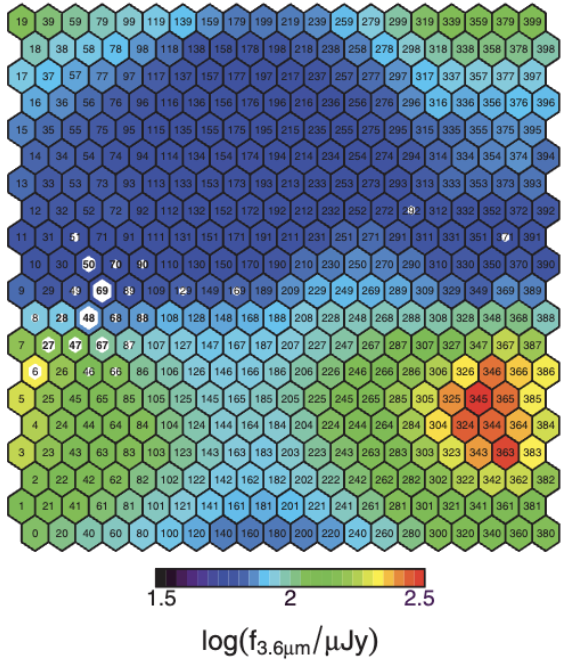
Group similar objects without being told what the groups are

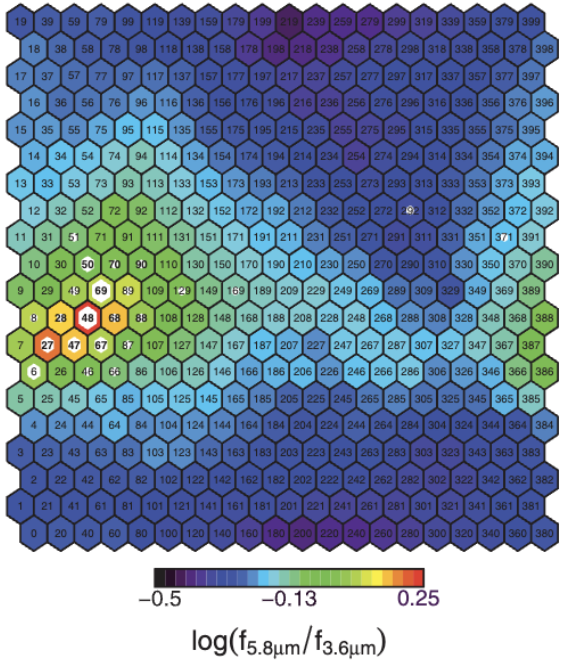
COMMON METHODS

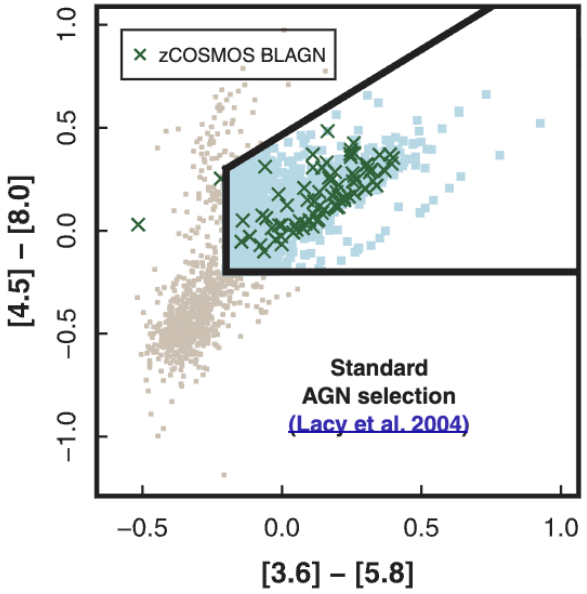
- K-means
- HDBSCAN
- Hierarchical clustering
- Self-organizing map

CAVEATS

- Finds groups even in data with no structure
- You must check whether the groups are physical







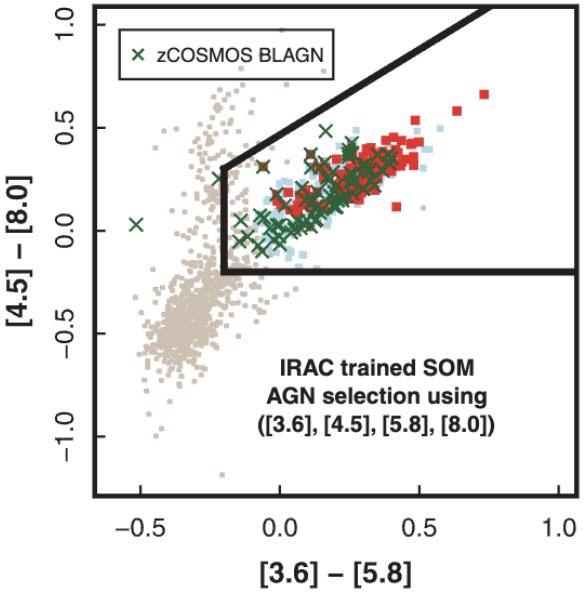


Figure from: Geach, J. E., 2012, MNRAS, 419(3), 2633-2645.

Dimensionality reduction

Unsupervised **Observations** **Modeling**

Describe the data with fewer variables while keeping what matters

COMMON METHODS

- Principal component analysis
- UMAP
- Autoencoders

CAVEATS

- The new axes rarely have physical meaning
- UMAP and similar methods distort distances

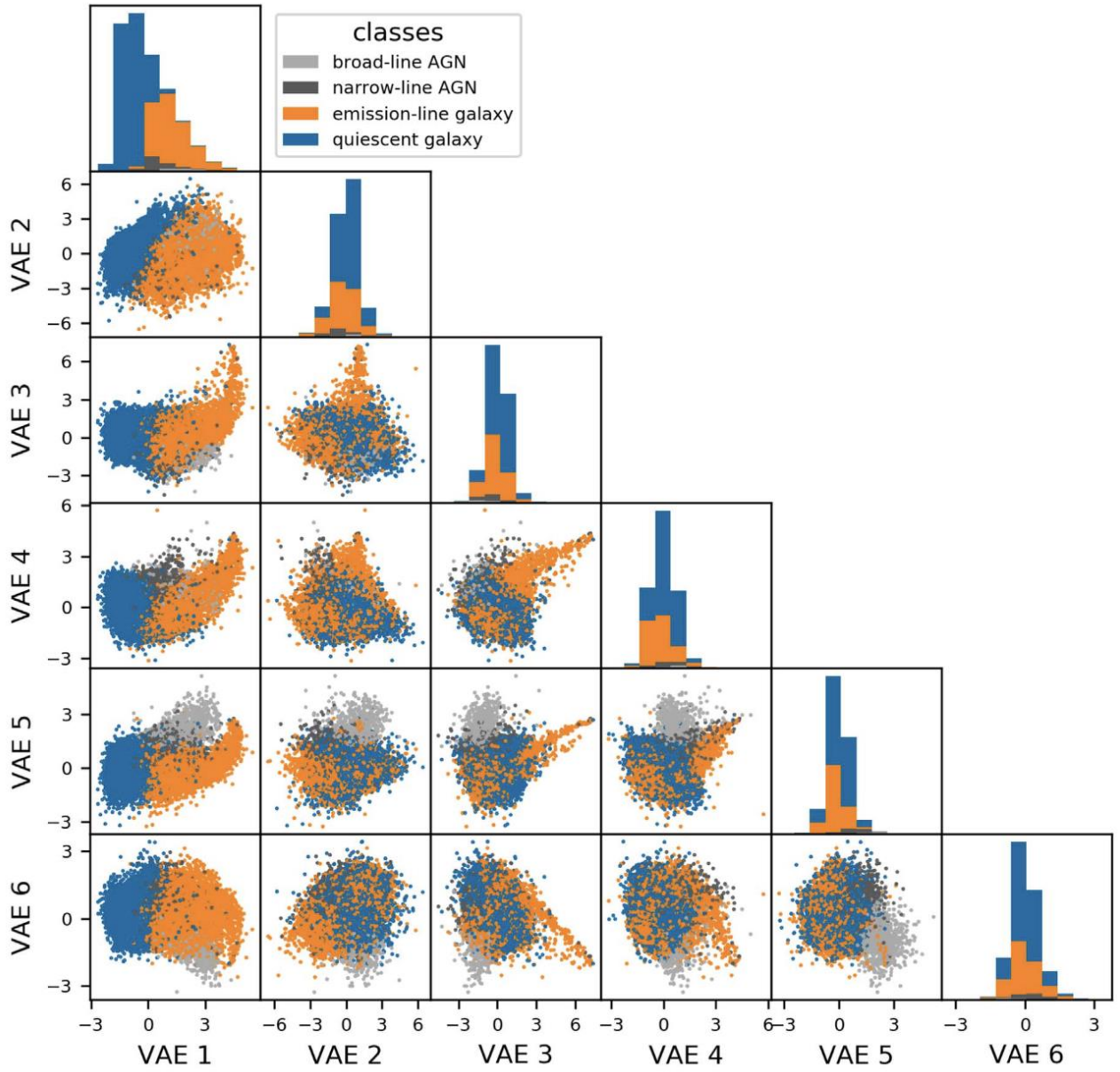


Figure from: Portillo, S. et al., 2020, The Astronomical Journal, 160(1), 45.

Neural networks, deep learning, CNNs

NEURAL NETWORK

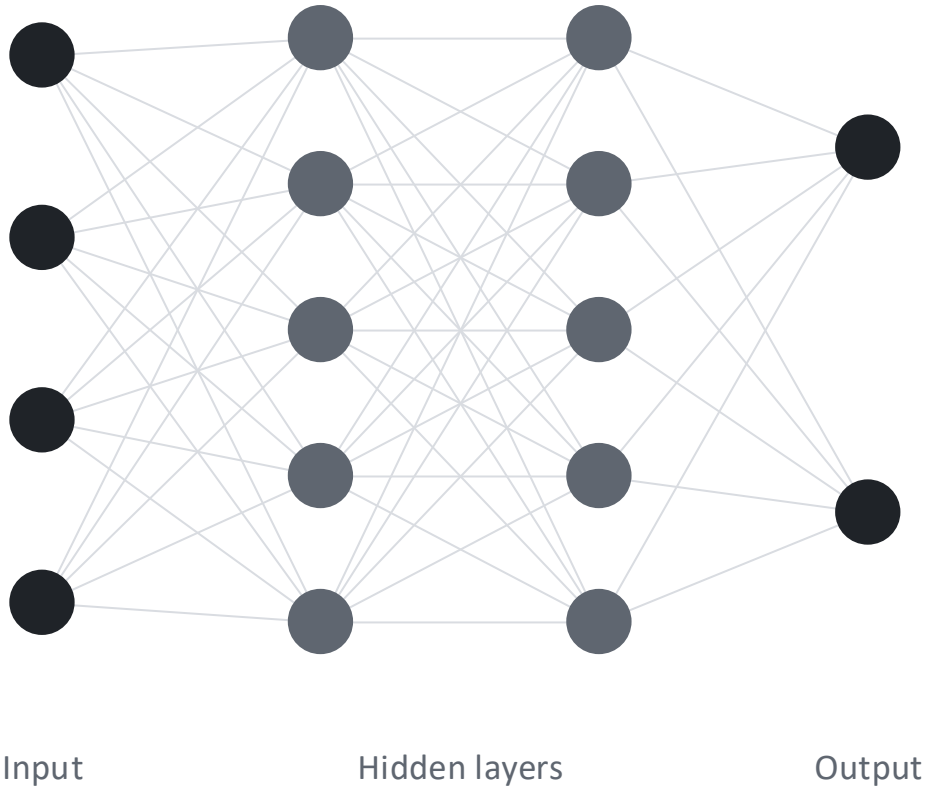
Layers of simple units, each a weighted sum passed through a nonlinear function

DEEP LEARNING

Neural networks with many hidden layers, so features build on features

CONVOLUTIONAL NEURAL NETWORK (CNN)

A deep network that scans images with small learned filters, spotting a pattern wherever it appears



Already in this lecture: the galaxy classifier is a CNN, and the spectra autoencoder is a neural network

Sources: Google Machine Learning Education, Machine Learning Roadmap

ML in Physics: Identify the type of problem

Observations

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A rule of thumb

A starting point, not a strict recipe

Model-based fitting

e.g., regression, MCMC. Reach for it when:

- You can write down a physical model and its likelihood
- You want the parameters themselves, with uncertainties
- Your data set is small or moderate

Machine learning

Reach for it when:

- There is no explicit model, or it is too slow to evaluate
- You want to sort, group, or predict across large data sets
- You have plenty of training data, real or simulated
- New data will resemble the training data

ML finds patterns and makes predictions within the range of what it has seen. It doesn't replace a physical model.

What’s next

In-class exercise

Lecture 2

- **The ML workflow**
  - Model selection
  - Data → preprocessing → splitting → training → evaluation → interpretation, repeat
- **Training and overfitting**
  - Evaluating models (confusion matrix, and sometimes you want different things)
- **Pitfalls in scientific ML**
  - Data leakage, selection effects, class imbalance, using it to predict things outside the scope of training data